MIL-T-82571A(OS)
31 January 1975
Superseding
MIL-T-82571(OS)
8 January 1969

MILITARY SPECIFICATION

TORPEDO TUBE, SURFACE VESSEL, MARK 32 MOD 9

(For Torpedoes Mk 44 Mod 1 and Mk 46 Mod 1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification establishes the requirements for the procurement of Surface Vessel Torpedo Tube Mk 32 Mod 9 (NAVORD LD 621171).

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-C-104 | Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted |
| MIL-P-116 | Preservation-Packaging Methods of |
| MIL-S-901 | Shock Tests, H. I. (High-Impact); Shipboard Machinery, Equipment and Systems; Requirements for |
| MIL-C-16173 | Corrosion Preventive Compound. Solvent Cutback, Cold Application |

FSC 1045

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-167 | Mechanical Vibrations of Shipboard Equipment |
| M1L-STD-81O | Environmental Test Methods |

DRAWINGS

Naval Sea Systems Command (Code Ident 10001)

| | |
|---|---|
| NAVORD LD 621171 | Torpedo Tube, Surface Vessel, Mk 32 Mod 9 (Assembly), Index to lists of Drawings |
| NAVORD LD 621172 | Torpedo Tube, Surface Vessel, Mk 32 Mod 9 Assembly |
| NAVORD Dwg 1448409 | Torpedo Tube, Surface Vessel, Mk 32 Mod 9 Assembly |
| NAVORD Dwg 1448418 | Torpedo Tube, Surface Vessel, Mk 32 Mod 9, Electrical Schematic |
| NAVORD Dwg 1448449 | Torpedo Tube, Surface Vessel, Mk 32 Mod 9, Electrical Schematic, External |
| NAVORD Dwg 1623922 | Pressure Gage |
| NAVORD Dwg 1984119 | Dummy Torpedo Mk 44 Mod 1 |
| NAVORD Dwg 2320787 | “GO” Gage, Barrel (Assembly) Torpedo Tube, Surface Vessel, Mk 32 Mods 0 through 9 |
| NAVORD Dwg 2652289 | Dummy Torpedo Mk 46 Mod 1 |
| NAVORD Dwg 3245120 | Pressure Fitting Adapter |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards Publications

| | |
|---|---|
| Handbook H28 | Screw-Thread Standards for the Federal Services |

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. Torpedo Tube Assembly Mk 32 Mod 9 is a stationary, pneumatic launching device capable of receiving, stowing, and launching Torpedoes Mk 44 and Mk 46. The tube assembly consists of two parallel barrels (Mk 10 Mods 10 and 11), one mounted above the other; four interchangeable breech mechanisms (Mk 1 Mod 1); and torpedo securing mechanisms. Each barrel is capable of being made ready to fire, fired, and returned to the safe or standby condition electrically from a remote location or manually at the tube. During manual firing a latch plate is swung open to provide automatic switching from normal to emergency gyro squib fire power and also to provide access to the firing solenoid plunger. The tube is equipped with an electrical junction box mounted on the left side of the barrel when viewed from the breech end. The junction box receives presetting functions from the fire control system and transmits them via electrical cabling to each barrel-housed torpedo. It also transmits the remote, safe, readying, and firing commands to the tubes and connects the electrical circuitry to the muzzle door heating units. Each muzzle door is pneumatically operated by ship's low pressure air and can be controlled remotely by electrical means or manually at the tube. A flexible charging hose is also supplied with the tube and is mounted on the deck house wall.

3.2 General requirements

3.2.1 Preproduction samples. Unless otherwise specified in the contract or order, a preproduction sample of Torpedo Tube Assembly Mk 32 Mod 9 shall be manufactured using the methods and procedures proposed for production. The sample will be tested as specified in Section 4 herein and is for the purpose-of determining, prior to production, that the contractor's production methods are capable of producing an item that meets the requirements of this specification and the design (See 6.2)

3.2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of Torpedo Tube Mk 32 Mod 9 shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in NAVORD LD 621172.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

3.2.3 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions of Torpedo Tube Mk 32 Mod 9 shall be strictly in accordance with the applicable drawings, specifications, and publications listed in NAVORD LD 621172.

* 3.2.4 Moisture, fungus, and corrosion. Torpedo Tube Mk 32 Mod 9 shall be rendered moisture, fungus, and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in NAVORD LD 621172.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in NAVORD LD 621172.

*

3.2.6 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

*

3.2.7 Conflicting requirements. In the event of conflict between the documents referenced herein and the contents of this specification, the contents of this specification shall apply.

3.2.8 Safety precautions . All standard safety precautions for high pressure air shall be strictly observed. No test shall be permitted to continue if in the opinion of the test engineer it is deemed unsafe. While conducting any of the firings, personnel shall stand clear of the breechward access opening and breech mechanism. Charging lines should be secured with safety chains when high pressure air charging operations are in progress.

3.3 Definitions. For the purpose of this specification the following definitions apply:

3.3.1 Torpedo tube assembly. The launching device in its assembled state.

3.3.2 Nonoperating condition. The condition of the tube when assembled, prepared, issued, transported, and handled.

3.3.3 Safe condition. The condition of the barrel assembly when the lever of the control valve operating mechanism is in the muzzleward position, the latch over the manual fire button of the solenoid valve is up and locked in position, and the muzzle door is closed and the manual knob linkage handle above the safe-ready solenoid valves is in the REMOTE position, as indicated on the support bracket.

3.3.4 Ready condition. That condition when the charged breech is installed and locked in position, the breakaway plug is mated to the torpedo, the breechward access cover is secured, and the muzzle door is open. The lever of the control valve operating mechanism is in the breechward position.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

3.3.5 Loaded condition. The condition of the tube when weapons are installed in the barrels.

3.4 Performance requirements and product characteristics. Torpedo Tube Assembly Mk 32 Mod 9 shall meet the following performance requirements and product characteristics:

3.4.1 Hydrostatic pressure, breech mechanisms. The breech mechanisms shall withstand a hydrostatic pressure of 4500 plus 20 minus O psig Without failure of any part.

* 3.4.2 High pressure pneumatic system pressure drop. The pneumatic system of each barrel of the torpedo tube, minus the breech mechanism, shall be attached to a 2000 psi air source through a 0.024 inch diameter orifice. With the securing mechanism removed and the manifold nipple blanked off, the pressure in the pneumatic system of each barrel shall stabilize at 1350 psig minimum. With the securing mechanism installed, the pressure shall. stabilize at 750 psig minimum. At the completion of this test there shall be no deformation or failure of any part of the pneumatic system.

3.4.3 Hydrostatic pressure, low Pressure pneumatic system. The low-pressure pneumatic system of each barrel shall withstand a hydrostatic pressure of 225 psig without failure of any part.

3.4.4 Breech mechanism air loss. When the four breech mechanisms are initially charged to 2000 psig, allowed to cool, and are then bled down to i600 plus or minus 10 psig, the air loss shall, after 24 hours, not exceed 5 percent air pressure.

* 3.4.5 Bore size of barrel. Each barrel shall pass a “GO” gage (NAVORD Dwg 2320787) with an axial force not to exceed 150 pounds.

3.4.6 Electrical circuitry. Circuitry shall be in accordance with NAVORD Drawing 1448418.

* 3.4.7 Dielectric strength. Electrical circuitry of the tube, with the exception of the muzzle door heater circuit, shall withstand the application of a 1000 volt ac potential between mutually insulated terminals and ground for one minute minimum without insulation breakdown. The applied rate shall not exceed 500 volts/second. The muzzle door heater circuit shall withstand the application of a 500 volt ac potential between input leads and ground for one minute minimum without insulation breakdown.

* 3.4.8 Insulation resistance. Immediately following the dielectric test, the insulation resistance between the hi-potted points in the muzzle door heater circuit shall be 1 megohm minimum at 500 volts dc, and in the remaining circuitry it shall be 3 megohms minimum at 500 volts dc.

* 3.4.9 Muzzle door heater operating current. The current drawn by each muzzle door heater circuit shall not exceed 8.4 amperes.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

* 3.4.10 Safe-ready solenoid operating current. The current drawn by each safe-ready solenoid circuit shall not exceed 0.4 ampere continuous (0.52 ampere inrush).

* 3.4.11 Firing solenoid operating current. The total current drawn by each firing solenoid shall not exceed 1.5 ampere continuous (1.95 inrush).

* 3.4.12 Total operating current. When all electrical components of one barrel of the torpedo tube are operating (with the exception of torpedo squib firing), the total maximum current drain shall be 10.5 ampere at 115 plus or minus 0.5 volts ac input.

* 3.4.13 Muzzle door heater resistance. The electrical resistance of each muzzle door heater circuit shall be 14.4 plus 1.6 or minus 0.7 ohms.

3.4.14 Muzzle door seal test. With force applied axially from the lip of the door farthest removed from the hinge, the axial force required to open the muzzle door, when no air is supplied to the pneumatic system, shall be 20 plus or minus 12 pounds.

3.4.15 Squib fire switch. When the latch plate is lowered, the switch plunger moves out forcefully, opening the normal squib fire circuit and closing the emergency squib fire circuit. Closing the emergency squib circuit automatically activates the course gyro in Torpedo Mk 46.

3.4.16 Ready light interlock circuits. The two ready light interlock circuits (one for each barrel, and each consisting of the muzzle door closed switch, ready light switch, and pressure switch) shall be given a continuity check during functional operation while the dry firing tests (4.4.5.22) and torpedo exit velocity tests (4.4.5.23) are being performed.

3.4.16.1 Control box ready light switch. When the switch actuator plate on the control box is placed in the ON position, the ready light switch contacts for that individual circuit shall close.

* 3.4.16.2 Pressure switch operation. The pressure switch contacts in each barrel shall close at 1275 plus or minus 75 psig and remain closed at pressures above this value. The pressure switch contacts shall open on falling pressure at 1275 plus or minus 75 psig and remain open at pressures below this value.

* 3.4.16.3 Muzzle door ready light switch. When the muzzle door is fully open (3.4.18) under 95 plus or minus 5 psig pressure, the ready light switch in the muzzle door shall close. The switch shall open when the muzzle door closes 6 degrees max from the fully open position.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

3.4.17 Breakaway plug puller mechanism. The lever of the breakaway plug puller mechanism shall be capable of being latched and unlatched by hand. The mechanism shall permit dry firings when in the latched position and shall remain in the latched position at the completion of these firings.

3.4.18 Muzzle door operation. With 95 plus or minus 5 psig supplied to the pneumatic circuit, and the doors operated manually and electrically, door closure shall be indicated by inability of a 0.030 feeler gage to penetrate the joint between the door face and the barrel ends. The full-open position of the door shall occur at 105 plus or minus 5 degrees. Door operation of both barrels must be checked.

3.4.19 Lighting interlock. The lighting interlock circuit consists of two muzzle door interlocks, two breechward access cover interlocks, and two breech mechanism locking handle interlocks. Should any sequence of operation allow light leakage from the tube, the interior deck house lighting will shut off.

3.4.20 Electrical continuity. Continuity shall be in accordance with NAVORD Drawing 1448418.

3.4.21 Lanyard retainer. The spring-loaded retainer shall be capable of accepting and securing a lanyard wire of 0.069 minimum diameter.

3.4.22 Dry firing. Each barrel of the torpedo tube assembly shall be capable of being fired without a torpedo by electrical and mechanical means, without damage, malfunction, or hindrance to subsequent firings with a torpedo.

3.4.23 Torpedo exit velocities. The exit velocities of each type of torpedo for various air flask pressures shall be in accordance with Schedule II, as specified in 4.4.5.23.

3.5 Environmental requirements

3.5.1 Vibration

3.5.1.1 Vibration, nonoperating. The torpedo tube assembly, less packing or packaging, shall be capable of meeting all quality conformance tests after being subjected to vibration tests of MIL-STD-167 for Type I vibrations in the three mutually perpendicular axes.

3.5.1.2 Vibration, standby condition. The torpedo tube assembly, loaded with dummy torpedoes and containing fully pressurized breech assemblies, shall be capable of meeting all quality conformance tests during and after being subjected to the vibration tests of MIL-STD-167 for Type I vibration in the three mutually perpendicular axes.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

3.5.2 Shock

3.5.2.1 Shock, nonoperating. The torpedo tube assembly, less packing or packaging, shall be capable of meeting all quality conformance tests after being subjected to the shock impulses of the magnitude and number specified in MIL-S-901 for medium weight equipment.

3.5.2.2 Shock, loaded condition. The torpedo tube, loaded with two dummy torpedoes, shall be capable of meeting all quality conformance tests after being subjected to the shock impulses of the magnitude and number specified in MIL-S-901 for medium weight equipment.

3.5.3 Temperature and humidity

* 3.5.3.1 Low temperature operation, loaded condition. The torpedo tube, less packaging and packing and in the loaded condition, shall be capable of withstanding the low temperature conditions specified in 4.5.3.1. While at the test temperature, the muzzle door heaters shall maintain the muzzle door operable and free from obstruction due to ice. At the conclusion of the test and while still at the test temperature, the torpedo tube shall show no damage and meet all the performance requirements and product characteristics of this specification.

* 3.5.3.2 High temperature operation. The torpedo tube, with fully pressurized breech assemblies, shall be capable of withstanding the high temperature and humidity conditions specified in 4.5.3.2. At the conclusion of the test and while still at the high temperature and humidity conditions, the torpedo tube shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

* 3.5.4 Marine atmosphere environment. The torpedo tube shall be capable of withstanding the simulated marine atmosphere or alternative test specified in 4.5.4 and thereafter shall show no evidence of corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.6 Cleanup. Prior to and after final assembly, all components shall be thoroughly cleaned of loose, spattered, or excessive solder, metal chips, and other foreign matter. Burrs and sharp edges, as well as rosin flash, shall be removed.

3.7 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to ensure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils,



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

marking of parts and assemblies, plating, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contractor order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of Torpedo Tube Mk 32 Mod 9 shall be classified as follows:

4.2.1 Preproduction and periodic production tests. Preproduction and periodic production tests are those which are accomplished on samples selected as specified in 4.3.1 and 4.3.3, which are representative of the production of the torpedo tube assembly after the award of the contract, to determine that the lot and production meet the requirements of this specification. These tests are detailed in 4.5. Acceptance shall be based on no defects in a sample. Failure of a sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.2.2 Quality conformance tests. Quality conformance tests are those performed on each torpedo tube assembly being submitted for acceptance under contract. These tests, detailed in 4.4, shall be perfomed by the manufacturer.

4.3 Sampling

* 4.3.1 Preproduction sample. A preproduction sample of one torpedo tube assembly, manufactured in accordance with 3.2.1 and preserved, packaged and packed as specified in the contract or order, shall be subjected to



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

the preproduction tests detailed in 4.5, at an activity designated by the procuring activity (see 6.2). Further production of the torpedo tube by the contractor prior to approval-of the preproduction sample shall be at the contractor's own risk. Approved preproduction samples shall be included as a part of the quantity specified for delivery in the contract or order.

* 4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

* 4.3.3 Periodic production sample. From each succeeding lot of 50 or less, a periodic production sample of one torpedo tube assembly shall be selected at random and submitted to an activity designated by the procuring activity (see 6.2) for periodic production inspection. The sample shall be examined for compliance with the requirements of the contract, specification and drawings and tested in accordance with 4.5. Failure of the sample to comply with any requirement shall result in the rejection of the lot as determined by the procuring activity (see 6.2).

4.3.4 Quality conformance sampling. The tests and examinations detailed in 4,4 shall be conducted on each torpedo tube assembly to be delivered; therefore, there shall be no sampling for quality conformance. Defective units shall be rejected.

4.4 Quality conformance inspection. Each Torpedo Tube Mk 32 Mod 9 to be delivered under contract shall be subjected to the tests and examinations specified in this section.

4.4.1 Test conditions. Unless otherwise specified, the torpedo tubes shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Test sequence. The sequence of testing procedures presented herein is suggested only. The tests may be performed in any order that will permit the most practical and economical use of manpower and equipment.

4.4.1.2 Temperature. Room ambient 65 degree'; Fahrenheit (F) to 95 degrees F (18 degrees centigrade (C) to 35 degrees C).

4.4.1.3 Altitude: Normal ground atmospheric pressure.

4.4.1.4 Vibration: None.



Source: https://assist.dla.mil -- Downloaded: 2026-04-23T07:15Z
Check the source to verify that this is the current version before use.

4.4.1.5 Humidity: Room ambient to 95 percent relative maximum.

* 4.4.2 Test results. When specified in the ccntract or order (See 6.2), test results shall be recorded, or verofocation noted, as required, opposite corresponding paragraph numbers on the Quality Conformance Inspection Data Sheet (Appendix) for each torpedo tube assembly inspected (See 5.4).

4.4.3 Acceptance criteria. Failure of Torpedo Tube Mk 32 Mod 9 to pass any examination or test of this section shall be cause for rejection.

4.4.4. Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, when definite requirements are not specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests. To perform the tests, the following items of test equipment, or equivalent, are required:

4.4.4.1 A means for producing hydrostatic pressure of 4500 psig.

4.4.4.2 A water pressure gage capable of reading 4500 psig with an accuracy of plus or minus 1/4 percent.

4.4.4.3 A 0-3000 psig filtered, clean, dry air source with particles not larger than 50 microns.

4.4.4.4 A 0-3000 psig pressure gage with an accuracy of plus or minus 1/4 percent at 2500 psig.

* 4.4.4.5 A “GO” gage for the barrel in accordance with NAVORD Dwg 2320787.

* 4.4.4.6 Standard high potential test equipment capable of outputs of 500 and 1000 volts, 60 to 1000 Hz.

* 4.4.4.7 Pressure fitting adapter (NAVORD Dwg 3245120).

4.4.4.8 A megohm bridge having a direct current test voltage of 500 *volts* and a range of 0.1 to 1,000,000 megohms.

* 4.4.4.9 An ohmmeter capable of measuring 13.7 to 16 ohms with an accuracy of plus or minus 5 percent of the reading.

4.4.4.10 A power supply capable of supplying at least 20 amps at 115 volts ac, 60 Hz, single phase, variable from 102 to 124 volts ac.

* 4.4.4.11 An ammeter capable of measuring 8.4 amps, 0.52 amp, and 1.95 amps with an accuracy of 0.375 amp.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

* 4.4.4.12 An ac voltmeter capable of measuring 102 to 124 volts ac with an accuracy of plus or minus 1.125 volts.

* 4.4.4.13 A $C0_2$ bottle of sufficient volume to cool the thermoswitches for the performance of the muzzle door heater resistance tests and the heater circuits operating current test.

* 4.4.4.14 A push scale with a memory pointer and capacity of at least 30 pounds with an accuracy of plus or minus 10 percent of the reading.

4.4.4.15 A means for measuring the angle of rotation of the muzzle door to the nearest. 1/2 degree.

* 4.4.4.16 A pull scale with a memory pointer and capacity of at least 155 pounds, with a maximum error of plus or minus 10 percent of the reading.

* 4.4.4.17 A test circuit capable of supplying 102 to 124 volts 60 Hz ac from the power supply of 4.4.4.10 to the firing solenoid for a period of 400 milliseconds maximum. Current capacity shall be at least 2 amperes.

* 4.4.4.18 DUmmy Torpedo MK 44 Mod 1 (NAVORD) Drawing 1984119) (weight: 433 plus or minus 7 pounds).

* 4.4.4.19 Dummy Torpedo Mk 46 Mod 1 (NAVORD Drawing 2652289) (weight: 508 plus or minus 8 pounds).

4.4.4.20 Launching ramp or track on which dummy torpedoes can be launched. The ramp or track shall include a means for decelerating and stopping the dummy torpedo on the track without damage.

4.4.4.21 Two infrared photoelectric control units suitable for use with electronic counter 4.4.4.22.

* 4.4.4.22 An electronic counter capable of recording up to 10,000 milliseconds in one-millisecond intervals with instant reset capabilities.

4.4.4.23 A continuity indicator.

4.4.4.24 A power supply of 28 volts or lower for use with the test circuit of 4.4.4.25.

4.4.4.25 A test circuit containing necessary circuitry to cut off electrical input power to the solenoid valve when the pressure switch cuts out. This circuit shall be capable of supplying 102 through 124 volts, 60 Hz ac from the power supply of 4.4.4.10 to the solenoid. Current capacity shall be 2 amperes maximum.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

* 4.4.4.26 A nitrogen bottle of sufficient volume to purge the breech mechanism air flasks and the low pressure pneumatic system.

* 4.4.4.27 A 0.024 plus or minus 0.0005 inch diameter orifice for the high pressure air source of 4.4.4.3.

4:4.5 Test procedures. The following tests shall be performed:

**WARNING**

When working with high pressure air, observe all safety precautions (see 3.2.8).

* 4.4.5.1 Hydrostatic pressure test, breech mechanisms. With the breech mechanisms removed from the tube assembly, remove Pressure Gage 1623922 and insert a plug in the void. the breech mechanisms shall be filled with water and pressurized to 4500 plus 20 minus O psig. This pressure shall be maintained for 10 minutes minimum without failure of any part. At the completion of this test, reduce the pressure to zero, drain the breech mechanism, and purge with dry nitrogen. Reinstall Pressure Gage 1623922.

**WARNING**

During the hydrostatic pressure tests, exercise care to stay clear of the safe-ready cylinder operating path.

* 4.4.5.2 High Pressure Pneumatic system Pressure drop test. 'Ibis test shall be performed on each barrel of the torpedo tube assembly. Remove the breech mechanism and securing mechanism from the barrel. Secure the nipple attached to the manifold and piping assembly to a shutoff valve or other suitable blanking device. Adjust the air source to 2000 plus O minus 20 psig and attach it through a 0.024 plus or minus 0.0005 inch diameter orifice to the barrel pneumatic system using a pressure fitting adapter (NAVORD Dwg 3245120). Actuate the firing solenoid valve and allow the pneumatic system line pressure to stabilize. The pressure as measured on the barrel side of the orifice shall stabilize at 1350 psig minimum. Repeat this test with the securing mechanism installed in the barrel. The pressure in the pneumatic system shall then stabilize at 750 psig minimum. At the conclusion of this test there shall be no deformation or failure of any part of the pneumatic system.

4.4.5.3 Hydrostatic pressure test, low Pressure pneumatic system. Fill the 100-150 psig system with water and pressurize to 225 psig. Maintain this pressure for 10 minutes with the barrel in the safe condition and for 10 minutes with the barrel in the ready condition. There shall be no failure of any part of the pneumatic system. After completion of the hydrostatic test, thoroughly purge water from the system with nitrogen.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

* 4.4.5.4 Breech mechanism air loss test. Gradually charge each breech mechanism (using the test equipment of 4.4.4.3 and 4.4.4.4) at a rate not exceeding 250 psig per minute to a pressure of 2000 psig pneumatic, and permit the mechanisms to cool for a period of 2 hours. The breech mechanisms shall then be bled to a pressure of 1600 plus or minus 10 psig. After a 24-hour period, observe the pressure, using gage 4.4.4.4. The pressure loss in each breech mechanism shall not exceed 5 percent.

* 4.4.5.5 Bore size of barrel test. The "GO" bore gage (4.4.4.5) shall pass through each barrel with an axial force not to exceed 150 pounds.

* 4.4.5.6 Ground circuit continuity. Continuity shall exist between terminal 4 of TB104 in the junction box and the junction box case as well as any of the two control box cases, breech interlock housings, and muzzle door mechanism housings.

* 4.4.5.7 Dielectric strength. For the purpose of this test the muzzle door heater switch shall be placed in the ON position and the muzzle door thermoswitches shall be cooled with $CO_2$ to close the contacts. The tube assembly shall withstand a potential of 1000 volts, 60 Hz ac applied between ground (terminal 4 of TB104 in the junction box) and all input terminals as shown on the external electrical schematic (NAVORD Dwg 1448449), with the exception of the muzzle door heater terminals, for one minute without evidence of breakdown. The input terminals may be connected together for this test. The muzzle door heaters shall withstand a potential of 500 volts 60 Hz ac applied between the input terminals and ground for a period of one minute with-':t evidence of breakdown.

* 4.4.5.8 Insulation resistance test. For the purpose of this test the the assembly shall be in the same condition as required for the dielectric strength test (4.4.5.7) with the exception that the muzzle door heater switch shall be in the OFF position. Immediately after the hipot test, the insulation resistance between the hipotted points shall be 3 megohms minimum at 500 vdc and the insulation resistance between terminals 10 and 11 of TB104 and ground (terminal 4 of TB104) shall be 1 megohm minimum at 500 vdc.

* 4.4.5.9 Muzzle door heater circuit operating current. With 115 plus or minus 0.5 volts 60 Hz ac applied to terminals 1 and 2 of TB1O4 in the junction box, the muzzle door heater switch in the ON position, and the thermoswitches of Barrel No. 1 closed by cooling with $C0_2$, measure the current. Permit the thermoswitch of Barrel No. 1 to return to the OPEN position and close the thermoswitches of Barrel No. 2 by cooling the switches; again measure the current. The current in each muzzle door heater circuit shall not exceed 8.4 amperes. If necessary, the muzzle door may be opened to permit access to the thermostats.



Source: https://assist.dla.mil -- Downloaded: 2026l0-23T07:15Z
Check the source to verify that this is the currentversion before use.

**CAUTION**

When testing solenoid operating currents (4.4.5.10 and 4.4.5.11), do not leave voltage across the solenoids for extended periods of time.

* 4.4.5.10 Safe-ready solenoid operating current. With 115 plus or minus 0.5 volts ac applied across terminals 6 and 9 of TB104, measure and record the current flow in Barrel No. 1. With 115 plus or minus 0.5 volts ac applied across terminals 2 and 5 of TB105, measure and record the current flow in Barrel No. 2. The total current drawn by the safe-ready solenoids in each barrel shall not exceed 0.4 ampere continuous (0.52 ampere inrush).

* 4.4.5.11 Firing solenoid operating current. With 115 plus or minus 0.5 volts ac applied across terminals 5 and 6 of TB104, measure and record the current flow in Barrel No. 1. With 115 plus or minus 0.5 volts ac applied across terminals 1 and 2 of TB105, measure and record the current flow in Barrel No. 2. The total current drawn by the solenoid in each barrel shall not exceed 1.95 amperes (inrush) and 1.5 amperes (continuous).

* 4.4.5.12 Total barrel operating current. The total operating current of each barrel, computed by adding the values recorded in 4.4.5.9, 4.4.5.10, and 4.4.5.11, shall not exceed 10.5 amperes.

* 4.4.5.13 Muzzle door heater resistance. With the thermoswitch cooled with C02 to close the heater circuit, measure the muzzle door heater resistance of Barrel No. 1 by disconnecting the leads to terminals 7 and 8 of TB102-1 in the control box and measuring the resistance across pins 7 and 8. Measure the heater resistance of Barrel No. 2 by cooling the thermoswitches, disconnecting the leads to terminals 7 and 8 of TB102-2, and measuring the resistance across pins 7 and 8. The resistance of the heater in each barrel shall be 14.4 plus 1.6 or minus 0.7 ohms.

4.4.5.14 Muzzle door seal test. With a pull scale engaged to the lip of the door farthest removed from the hinge, apply sufficient force in an axial direction to open the muzzle doors. The force required to open each door shall be 20 plus or minus 12 pounds.

4.4.5.15 Squib fire switch. With the control box latch plates in the locked position, continuity shall exist across terminals 8 and 9 of TB101-1 in the control box of Barrel No. 1 and across terminals 8 and 9 of TB101-2 in the control box of Barrel No. 2. Release the latch on the control box of each barrel and observe that the switch plunger moves out forcefully; also verify that continuity exists across terminals 8 and 10 of TB101-1 and terminals 8 and 10 of TB101-2.

4.4.5.16 Ready light interlock circuit. With the interlock switches of 4.4.5.16.1, 4.4.5.16.2, and 4.4.5.16.3 closed, continuity shall exist between terminals 7 and 8 of TB104 for Barrel No. 1 and between terminals 3 and 4 of



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

TB105 for Barrel No. 2. Continuity will be broken in the individual circuit with the opening of any one of the above three interlocks.

4.4.5.16.1 Control box ready light switch. With the switch actuator plate in the OFF position, the control box ready light switch will open for each barrel circuit. With the plate in the ON position the switch will close. Continuity shall exist as called for in 4.4.5.16.

* 4.4.5.16.2 Pressure switch. Gradually apply pressure to the air flask at a rate not exceeding 250 psig per minute until the switch closes. The pressure switch in each barrel shall close at 1275 plus or minus 75 psig and remain closed at pressures above this value. Gradually decrease pressure until the pressure switch opens. The pressure switch in each barrel shall open at 1200 psig minimum and remain open at pressures below this value. Continuity shall exist as called for in 4.4.5.16.

* 4.4.5.16.3 Muzzle door ready light switch. With 95 plus or minus 5 psig pressure applied to the low pressure air system, manually rotate the safe-ready solenoid linkage handle to the MANUAL position. The muzzle door ready light switch will close with the muzzle door fully open as indicated in 3.4.18. Return the solenoid linkage handle to the REMOTE position. When the door closes 6 degrees maximum from the fully open position, the switch will open. Continuity shall exist as called for in 4.4.5.16.

4.4.5.17 Breakaway plug puller mechanism. Verify that, prior to all firings, the lever of the breakaway plug puller mechanism in each barrel can be latched and unlatched by hand without excessive force.

* 4.4.5.18 Muzzle door operation. With 95 plus or minus 5 psig air supplied to the pneumatic system, open and close the muzzle door by rotating the linkage handle above the safe-ready solenoid valves to the MANUAL position and then back to the REMOTE position. With the safe-ready solenoids in the REMOTE position, open and close the door by applying and removing 115 volts ac to the safe-ready solenoid valves.

4.4.5.18.1 The doors when closed shall not accept a 0.030 feeler gage in the joint between the door and the barrel ends.

4.4.5.18.2 The doors when open shall have rotated 105 degrees plus or minus 5 degrees from the closed position.

* 4.4.5.19 Lighting interlock circuit. With the breech mechanisms of both barrels locked in place and the access covers secured, apply 95 plus or minus 5 psig pneumatic pressure to the low pressure system and close both muzzle doors by placing the safe-ready solenoid valve linkage handle in the REMOTE position.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

4.4.5.19.1 Continuity. Continuity shall exist across terminals 8 and 9 of TB105 in the tube junction box.

4.4.5.19.2 Muzzle door switches. Open both muzzle doors by manually turning the linkage handle above the safe-ready solenoid valves to the MANUAL position. Continuity shall exist between terminals 8 and 9 of TB105.

4.4.5.19.3 Access opening switches. With the muzzle doors open, open the access cover of each barrel. Opening of the access cover of either barrel shall break continuity between terminals 8 and 9 of TB105. Close access covers.

4.4.5.19.4 Breech mechanism locking handle switch. Rotate the breech locking ring of each barrel to the unlocked position, then back to the locked position. Unlocking the ring on each barrel shall break continuity between terminals 8 and 9 of TB105.

4.4.5.20 Continuity. Continuity shall be in accordance with NAVORD Drawing 1448418.

4.4.5.21 Lanyard retainer. The lanyard retainer plunger shall be depressed by hand and a loop of 0.069 minimum diameter wire shall be assembled to each retainer. The plungers shall depress by hand pressure and the retainers shall accept and secure the loop of 0.069 minimum diameter wire.

4.4.5.22 Dry firing tests. Each breech mechanism shall be charged to Pressure in accordance with Schedule I at a rate not exceeding 250 psig per minute. The low pressure system shall be energized with 95 plus or minus 5 psig pneumatic pressure. The control valve operating lever shall be placed in the ready condition by manual or electrical operation of the safe-ready solenoid valves as indicated in Schedule I. The plug puller mechanism shall be placed in the latched position. Each barrel shall be fired a total of five times without a torpedo in place. The input voltage to the firing solenoid and safe-ready solenoids shall be varied in accordance with Schedule 1. The application of test voltage during firings shall be regulated by the test circuit of 4.4.4.17, which shall limit firing potential duration to 400 milliseconds maximum. After each firing the breech mechanism shall be rotated to ensure that the breech mechanisms are interchangeable.

4.4.5.22.1 Safe-ready solenoid valve test. The safe-ready solenoids in each barrel shall be operated manually and electrically in accordance with Schedule I. Proper operation shall be indicated by door and control valve lever operation.

4.4.5.22.2 Safe-ready cylinder operation. The safe-ready cylinder shall operate in such a manner that the control valve operating plunger shall not engage the control valve until the muzzle door is within 5 degrees of the full open position.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

4.4.5.22.3 Control valve operating lever. During all dry firing tests observe that there is no leakage or premature air escape while the control valve is in the safe condition.

4.4.5.22.4 Ready light circuit operation. During the dry firing tests, verify that the ready light circuit condition is in accordance with Schedule I when the control valve operating lever is in the READY condition. Measurements should be made across terminals 7 and 8 of TB104 for Barrel No. 1 and across terminals 3 and 4 of TB105 for Barrel No. 2.

4.4.5.22.5 Breakaway plug mechanism. After each dry firing, the breakaway plug mechanism shall be in the latched position and shall be capable of being unlatched by hand.

4.4.5.22.6 Electrical firing test. Each barrel electrically fired shall fire with only one 400 millisecond application of test potential. Two firings shall be made at 102.5 plus or minus 0.5 volts ac and two firings at 123.5 plus or minus 0.5 volts ac.

* SCHEDULE I - DRY FIRING TESTS

NOTE: Each barrel is to be fired five times as outlined below.

| | Firing Pressure | AC Voltage To Solenoid | Readying Method | Firing Method |
|---|---|---|---|---|
| 1 | 1200 | 123.5 ± 0.5 | Remote | Manual** |
| 2 | 1350 | 123.5 ± 0.5 | Manual | Electrical |
| 3 | 2000 | 123.5 ± 0.5 | Remote | Electrical |
| 4 | 1350 | 102.5 ± 0.5 | Manual | Electrical |
| 5 | 2000 | 102.5 ± 0.5 | Remote | Electrical |

**Indicates firing is to be attmpted electrically prior to manual firing. This is to check pressure switch control of firing circuit. The barrel cannot be fired electrically unless the ready light is on.

Ensure that the applicable barrel is secured at proper attitude to be compatible with track.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

4.4.5.22.7 Observation for damage. At the completion of the dry firing test, there shall be no damage sustained by the torpedo or torpedo tube assembly.

* 4.4.5.23 Torpedo exit velocity tests. The two photoelectric cell units of 4.4.4.21 shall be spaced three feet apart alongside the track or ramp. The first set of cells shall be located the length of a torpedo plus or minus 0.5 foot from the muzzle end of the barrel. The photoelectric cells shall be attached to the electronic counter 4.4.4.22 so that when the first beam is broken by the nose of the torpedo the counter starts and when the second beam is broken by the nose of the torpedo, the counter stops. The average velocity shall then be calculated by the formula v=d/t where d=distance in feet, V= average velocity, and t=time in seconds. Each breech mechanism shall be charged at a rate not exceeding 250 psig per minute. The breech mechanism should be charged to a pressure of 200 psig greater than those pressures specified in Schedule II and allowed to cool for at least 5 minutes prior to being bled and readied for a firing in accordance with Schedule II. The minimum exit velocity shall be met in all firings.

Should this method of measuring velocity be impractical or difficult for the manufacturer, an alternate means, proposed by the manufacturer and approved by the contracting activity, shall be considered a valid test.

4.4.5.23.1 Safe-ready cylinder operation. The safe-ready cylinder shall operate- in such a manner that the control valve operating plunger shall not engage the control valve until the muzzle door is within 5 degrees of the full open position.

4.4.5.23.2 Control valve operating lever. During all tests observe that when the control valve operating lever is in the SAFE condition there is no leakage or premature air escape.

4.4.5.23.3 Ready light circuit operation. During the firing tests, verify that continuity of the ready light circuit exists in accordance with Schedule II when the ready light switch is in the ON position, the control valve operating lever is in the READY condition, and the muzzle door is open.

4.4.5.23.4 Breakaway plug mechanism

* 4.4.5.23.4.1 Prior to each firing the breakaway plug mechanism shall be placed in the unlatched position and the breakaway plug attached to the torpedo receptacle.

* 4.4.5.23.4.2 After each firing the plug puller mechanism lever shall be in the latched position and shall be capable of being unlatched by hand.

* 4,4.5.23.4.3 After each firing the breakaway plug shall be inspected for damage.



Source: https://assist.dla.mil -- Downloaded: 2026l0-23T07:15Z
Check the source to verify that this is the currentversion before use.

* 4.4.5.23.5 Electrical firing test. Each electrical firing test shall be made with only one 400 millisecond application of test potential. The test potential for these tests shall be 115 plus or minus 0.5 volts ac.

4.4.5.23.6 Manual firing. During the manual firing tests, measure the force required to actuate the solenoid valve with the push scale. The force required shall be 25 pounds maximum.

* SCHEDULE II - EXIT VELOCITY TESTS

NOTE: A Torpedo Mk 44 Mod 1, weighing 433 plus or minus 7 pounds, is to be fired three consecutive times as outlined below.

| | Firing Pressure (psig) | Firing Method | Min. Exit Velocity (ft/sec) | Voltage Requirement (volts) |
|---|---|---|---|---|
| 1 | 1200 | Manual | 40 | 115 |
| 2 | 1350 | Electrical | 40 | 115 |
| 3 | 1500 | Electrical | 40 | 115 |

NOTE: A torpedo Mk 46 Mod 1, weighing 508 plus or minus 8 pounds, is to be fired three consecutive times as outlined below.

| | Firing Pressure (psig) | Firing Method | Min. Exit Velocity (ft/sec) | Voltage Requirement (Volts) |
|---|---|---|---|---|
| 1 | 1200 | Manual | 40 | 115 |
| 2 | 1350 | Electrical | 40 | 115 |
| 3 | 1500 | Electrical | 40 | 115 |

4.4.5.23.7 Observation for damage. At the completion of velocity tests, there shall be no damage to the torpedo or torpedo tube assembly.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

* 4.4.5.24 Packaging, packing, and marking. The packaging, packing, and marking of the torpedo tube assembly shall be examined to ensure that it complies fully with the requirements of Section 5.

4.5 Preproduction and Periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Failure to meet any requirement specified herein shall be considered cause for rejection of the lot represented and units tested.

4.5.1 Vibration tests

4.5.1.1 Vibration, nonoperating. The torpedo tube, secured to the mounting bracket of the testing machine in the same manner that it will be secured on shipboard, shall be subjected to the exploratory vibration test, variable frequency test, and endurance test as specified in MIL-STD-167 for Type I vibrations.

* 4.5.1.2 Vibration, standby condition. The top barrel of the torpedo tube shall be loaded with a dummy Mk 46 Mod 1 Torpedo, and the bottom barrel shall be loaded with a dummy Mk 44 Mod 1 Torpedo. Both breech assemblies shall be pressurized to 2000 psig. The torpedo tube, loaded as specified and secured to the mounting bracket of the testing machine in the same manner that it will be secured on shipboard, shall be subjected to the exploratory vibration test, variable frequency test, and endurance test as specified in MIL-STD-167 for Type I vibrations.

4.5.2 Shock tests

* 4.5.2.1 Shock, nonoperating. The torpedo tube assembly shall be attached to the anvil table of the shock testing machine by means of a standard mounting platform and a 30 degree mounting fixture as specified in MIL-S-901 for medium-weight equipment. The torpedo tube shall be oriented on the 30 degree mounting fixture such that the barrel center lines are parallel to the horizontal plane. In each mounting condition the torpedo tube shall be subjected to Type A shock tests as specified in MIL-S-901 for medium-weight equipment.

4.5.2.2 Shock, loaded condition. The torpedo tube assembly shall be attached to the anvil table of the shock testing machine by means of a standard mounting platform and a 30 degree mounting fixture as specified in MIL-S-901 for medium-weight equipment. The lower barrel of the torpedo tube shall be loaded with a dummy Torpedo Mk 44 Mod 1 and the top barrel shall be loaded with a dummy Torpedo Mk 46 Mod 1. The torpedo tube shall be oriented on the 30 degree mounting fixture such that the barrel centerlines are 30 degrees from the horizontal plane. In each mounting condition the torpedo tube shall be subjected to Type A shock tests as specified in MIL-S-901 for medium-weight equipment.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

4.5.3 Temperature and humidity tests

* 4.5.3.1 Low temperature tests, loaded condition. Two dummy torpedoes shall be loaded in the torpedo tube. One dummy torpedo shall be of the Mk 44 Mod 1 type and the other dumny torpedo shall be of the Mk 46 Mod 1 type. Thermocouples shall be installed on the surface of the torpedoes in the area of contact between the torpedo and the barrel quadrants. Provisions shall be made for reading the sensed temperatures. The muzzle door end of the torpedo tube shall then be placed in a test chamber and the muzzle door heaters turned on. The temperature of the test chamber shall be lowered to -29 ± 3 degrees C (-20 - 5 degrees F) and maintained at this temperature for a period of 48 plus or minus 5 hours. At the end of 24 hours in the test chamber, the muzzle doors of the torpedo tube shall be liberally sprayed with fresh water having a temperature of plus 32 degrees F. At the end of 48 plus or minus 5 hours, the muzzle door assembly shall be checked for operability and the complete torpedo tube shall be performance checked. The temperatures recorded for the torpedoes shall at no the exceed 37.5°C (lOO°F) or be less than 0.5°C (33°F).

* 4.5.3.2 High temperature and humidity tests. The muzzle door of the torpedo tube assembly forward of the bulkhead adapter ring shall be placed within a test chamber and subjected to temperature and humidity cycling as specified in MIL-STD-81O Method 507, Procedure IV with the following exceptions:

(a) Referenced measurements of steps 4, 5, 6, and 8 may be deleted.

(b) The torpedo tube assembly shall be performance checked during the nigh temperature portion of the fifth cycle of step 5 prior to decreasing to 30°c (86°F).

4.5.4 Marine environment

* 4.5.4.1 Marine atmosphere environment test. The muzzle door forward of the bulkhead adapter ring shall be exposed at dockside or on the weather deck of a ship for 60 plus or minus 5 days. The access covers and muzzle doors shall be opened for a period of 10 plus or minus 1 days. The assembly shall be protected from rain or snow by an appropriate shelter that permits free circulation of air. At the completion of this test, there shall be no evidence of corrosion.

* 4.5.4.2 Alternative test for marine atmosphere. Subject the muzzle door of the torpedo tube forward of the bulkhead adapter ring to the Salt Fog test of MIL-STD-81O, Method 509, using a 20 percent salt solution. The solution shall be prepared by dissolving 20 plus or minus 2 parts by weight of salt in 80 parts by weight of distilled water. The solution shall be adjusted to and maintained at a specific gravity between 1.126 and 1.157. At the completion of this test there shall be no evidence of corrosion.



Source: https://assist.dla.mil -- Downloaded: 2026l0-23T07:15Z
Check the source to verify that this is the current version before use.

4.5.5 During and after performance of the above environmental tests, no damage shall occur which could affect proper operation or function.

4.5.6 At the completion of the above environmental tests, the torpedo tube shall meet all the quality conformance rests of 4.4.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning. Unless otherwise specified, the torpedo tube assembly shall be cleaned in accordance with Method C-1 of MIL-P-116.

5.1.1.2 Drying. Unless otherwise specified, the torpedo tube shall be dried by one or more of the procedures, as deemed suitable, of MIL-P-116.

5.1.1.3 Preservation. The torpedo tube shall be placed in the nonoperating condition. Immediately after cleaning and drying, the external unpainted metal surfaces which are subject to corrosion shall be perserved by the application of preservative compound conforming to the requirements of MIL-C-16173, Grade 2.

5.1.1.4 Unit packaging. Not applicable.

5.1.2 Level B. Not applicable.

5.1.3 Level C. Cleaning, drying, preservation, and packaging may be in accordance with the supplier's commercial practice if such practice has been approved by the procuring activity.

5.2 Packing

5.2.1 Level A

* 5.2.1.1 Exterior containers. The torpedo tube assembly, in the nonoperating condition, shall be packed in a nailed wood, skid-mounted crate with rubbing strips in accordance with MIL-C-104 for Class 1, Style A, Type I crates. The base of the torpedo tube shall be rigidly fastened to the load bearing members of the skid base and the tube barrels shall be braced and cushioned to prevent movement.

5.2.2 Level B. Not applicable.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. The torpedo tube assembly, in the nonoperating condition, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall comply with the carrier rules and regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Special markings. None, unless otherwise specified.

5.3.2 Normal markings. In addition to the markings required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129, including serial numbers.

* 5.4 lnspection data sheets. When Quality Conformance Inspection Data sheets are specified in the contract or order (See 6.2.2) for completion in accordance with 4.4.2, one data sheet (or copy thereof), bearing recorded data, shall be shipped with each torpedo tube assembly inspected. The torpedo tube assembly equipment serial number shall correspond to the serial number recorded on the accompanying data sheet.

6. NOTES

6.1 Intended use. Torpedo Tube Assembly Mk 32 Mod 9 is intended for use in an amidship deckhouse on the 01 level of surface vessels. It will be used with Attack Console Mk 53 Mod O of Fire Control System Mk 114.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Required level of preservation and packaging. (see 5.1)

(c) Required level of packing. (see 5.2)

(d] Assigned activity for preproduction and periodic production tests. (4.2.1)

(e) Any exceptions to this specification. (3.2.1, 3.2.2, 3.2.6, 4.4.1 & 5.3)

(f) That all additional samples required shall be provided and inspected at the expense of the supplier, when such additional samples are required because of failure of any sample(s) to pass the prescribed inspections.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in MIL-P-116.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

| Paragraph | Data Requirements | Applicable DIDS |
|---|---|---|
| 4.4.2 | Test results | UD1-E-20602 |

DIDS (Data Item Description/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDS when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations . Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian
Navy- OS

Preparing Activity
Navy - OS
Project No. 1045-NO1O



Source: https://assist.dla.mil -- Downloaded: 2026t0-23T07:15Z
Check the source to verify that this is the currentversion before use.

(This page intentionally left blank)



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

DATA SHEET

QUALITY CONFORMANCE INSPECTION

TESTS AND EXAMINATIONS

FOR

TORPEDO TUBE MARK 32 MOD 9

(LD 621171)

EQUIPMENT STATUS

| CONTRACTOR CERTIFICATION | | GOVERNMENT ACCEPTANCE | |
|---|---|---|---|
| Contractor | | Date of Acceptance | |
| Contract | | Accepted by | |
| Equipment Ser. No. | | Title | |
| Insp. Req. Satisfied | | Agency | |

ORDALT STATUS

| ORDALT NO. | INSTALLED | ORDALT NO. | INSTALLED |
|---|---|---|---|
| | | | |
| | | | |



Source: https://assist.dla.mil -- Downloaded: 202640-23T07:15Z
Check the source to verify that this is the current version before use.

DATA SHEET

QUALITY CONFORMANCE INSPECTION

TORPEDO TUBE MARK 32 MOD 9, Equipment Serial No._____

10. SCOPE

10.1 This appendix contains data sheets to record the results of the quality conformance tests and examinations of 4.4 of this specification. When completed, with authorized certification and acceptance signatures affixed, this report form or copies thereof shall accompany the Torpedo Tube Assembly bearing the recorded equipment serial number.

20. GENERAL INSTRUCTIONS

20.1 Test sequence. When the contract or *order* specifies use of this appendix, the Quality Conformance Inspection tests of 4.4 shall be completed and the results recorded or verified in these data sheets prior to acceptance of the Torpedo Tube Assembly. Unless otherwise noted, the sequence of testing procedures presented herein is suggested only. The procedures may be performed in any order which will permit the most practical and economical use of manpower and equipment.

20.2 Recording of results. Where space is provided in the "Reading" column of the Quality Conformance Test data sheets, the value or specific reading obtained shall be recorded and the initials of the individual performing the inspection shall be inserted above the value observed and recorded. If limits are not stated, the individual performing the test or inspection shall initial in the space provided in the "Reading" column indicating that the requirements have been met satisfactorily.



Source: https://assist.dla.mil -- Downloaded: 2026-04-23T07:15Z
Check the source to verify that this is the current version before use.

| | Reference | Limits | Reading |
|---|---|---|---|
| 30. ACCEPTANCE TEST AND EXAMINATION | | | |
| 30.1 Test equipment | 4.4.4 | In calibration | _____ |
| 30.2 Test procedure | | | |
| 30.2.1 Hydrostatic pressure (breech mechanisms) | 4.4.5.1 | | |
| 4500 psig hydrostatic pressure, Breech Mechanism No. 1 | | Verified | _____ |
| 4500 psig hydrostatic pressure, Breech Mechanism No. 2 | | Verified | _____ |
| 4500 psig hydrostatic pressure, Breech mechanism No. 3 | | Verified | _____ |
| 4500 psig hydrostatic pressure, Breech Mechanism No. 4 | | Verified | _____ |
| * 30.2.2 High pressure pneumatic system pressure drop | 4.4.5.2 | | |
| Securing mechanism removed | | | |
| Barrel No. 1 | | 1350 psig min | _____ |
| Barrel No. 2 | | 1350 psig min | _____ |
| Securing mechanism installed | | | |
| Barrel No. 1 | | 750 psig min | _____ |
| Barrel No. 2 | | 750 psig min | _____ |
| 30.2.3 Hydrostatic pressure (low pressure system) | 4.4.5.3 | | |
| 225 psig hydrostatic pressure, low pressure pneumatic system, Barrel No. 1 | | Verified | _____ |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

| | Reference | Limits | Reading |
|---|---|---|---|
| 225 psig hydrostatic pressure, low pressure pneumatic system, Barrel No. 2 | | Verified | _____ |
| 30.2.4 Breech mechanism air loss | 4.4.5.4 | | |
| Breech Mechanism No. 1 air loss | | 80.8 lbs max pressure loss | _____ |
| Breech Mechanism No. 2 air loss | | 80.8 lbs max pressure loss | _____ |
| Breech Mechanism No. 3 air loss | | 80.8 lbs max pressure loss | _____ |
| Breech Mechanism No. 4 air loss | | 80.8 lbs max pressure loss | _____ |
| * 30.2.5 Bore size | 4.4.5.5 | | |
| Barrel No. 1 | | 150 Ibs max | _____ |
| Barrel No. 2 | | 150 lbs max | _____ |
| 30.2.6 Ground circuit continuity | 4.4.5.6 | Verified | _____ |
| 30.2.7 Dielectric strength | 4.4.5.7 | Verified | _____ |
| * 30.2.8 Insulation resistance | 4.4.5.8 | | |
| Heater switch in OFF position | | 3 megohms min | _____ |
| Heater circuit | | 1 megohm min | _____ |
| * 30.2.9 Muzzle door heater circuit operating current | 4.4.5.9 | | |
| Barrel No. 1 | | 8.4 amps max | _____ |
| Barrel No. 2 | | 8.4 amps max | _____ |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

| | Reference | Limits | Reading |
|---|---|---|---|
| * 30.2.10 Safe-ready solenoid operating current | 4.4.5.10 | | |
| Barrel No. 1 | | 0.4 amp max | _____ |
| Barrel No. 2 | | 0.4 amp max | _____ |
| * 30.2.11 Firing solenoid operating current | 4.4.5.11 | | |
| Barrel No. 1 | | 1.5 amp max | _____ |
| Barrel No. 2 | | 1.5 amp max | _____ |
| * 30.2.12 Total barrel operating current | 4.4.5.12 | | |
| Barrel No. 1 | | 10.5 amps max | _____ |
| Barrel No. 2 | | 10.5 amps max | _____ |
| * 30.2.13 Muzzle door heater resistance | 4.4.5.13 | | |
| Barrel No. 1 | | 14.4 + 1.6 or -0.7 ohms | _____ |
| Barrel No. 2 | | 14.4 + 1.6 or -0.7 ohms | _____ |
| 30.2.14 Muzzle door seal test | 4.4.5.14 | | |
| Barrel No. 1 | | 20 ± 12 lbs | _____ |
| Barrel No. 2 | | 20 ± 12 lbs | _____ |
| 30.2.15 Squib fire switch | 4.4.5.15 | | |
| Barrel No. 1 | | Operates properly | _____ |
| Barrel No. 2 | | Operates properly | _____ |
| * 30.2.16 Ready light interlock circuit | 4.4.5.16 | | |
| Barrel No. 1 | | Continuity exists | _____ |
| Barrel No. 2 | | Continuity exists | _____ |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

| | Reference | Limits | Reading |
|---|---|---|---|
| Control box switch | 4.4.5.16.1 | | |
| Barrel No. 1 | | | |
| ON position | | Switch closes | _____ |
| OFF position | | Switch opens | _____ |
| Switch open | | Continuity broken | _____ |
| Barrel No. 2 | | | |
| ON position | | Switch closes | _____ |
| OFF position | | Switch opens | _____ |
| Switch open | | Continuity broken | _____ |
| Pressure switch | 4.4.5.16.2 | | |
| Barrel No. 1 | | | |
| 1275 ± 75 psig | | Switch closes | _____ |
| 1200 psig min | | Switch opens | _____ |
| Switch open | | Continuity broken | _____ |
| Barrel No. 2 | | | |
| 1275 ± 75 psig | | Switch closes | _____ |
| 1200 psig min | | Switch opens | _____ |
| Switch open | | Continuity broken | _____ |
| Muzzle door switch | 4.4.5.16.3 | | |
| Barrel No. 1 | | | |
| Door full open | | Switch closes | _____ |
| Full open less 6° | | Switch opens | _____ |
| Switch open | | Continuity broken | _____ |
| Barrel No. 2 | | | |
| Door full open | | Switch closes | _____ |
| Full open less 6° | | Switch opens | _____ |
| Switch open | | Continuity broken | _____ |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

| | Reference | Limits | Reading |
|---|---|---|---|
| 30.2.17 Breakaway plug puller mechanism | 4.4.5.17 | | |
| Barrel No. 1 | | Hand latchable<br>Hand unmatchable | _____<br>_____ |
| Barrel No. 2 | | Hand latchable<br>Hand unmatchable | _____<br>_____ |
| 30.2.18 Muzzle door operation | 4.4.5.18 | | |
| Barrel No. 1 (NO-GO test) | | Verified | _____ |
| Barrel No. 1 (open position) | | 105° ± 5° | _____ |
| Barrel No. 2 (NO-GO test) | | Verified | _____ |
| Barrel No. 2 (open position) | | 105° ± 5° | _____ |
| 30.2.19 Lighting interlock circuit | 4.4.5.19 | Verified | _____ |
| Continuity | 4.4.5.19.1 | Verified | _____ |
| Muzzle door switch | 4.4.5.19.2 | Verified | _____ |
| Access opening switches | 4.4.5.19.3 | verified | _____ |
| Breech interlock switch | 4.4.5.19.4 | Verified | _____ |
| 30.2.20 Continuity | 4.4.5.20 | Verified | _____ |
| 30.2.21 Lanyard retainer | 4.4.5.21 | | |
| Barrel NO. 1 | | Verified | _____ |
| Barrel No. 2 | | Verified | _____ |
| 30.2.22 Dry firing test | 4.4.5.22 | Breech mechanism functioned in all barrels | _____ |
| Safe-ready solenoid valve | 4.4.5.22.1 | | |
| Barrel No. 1 | | Verified | _____ |
| Barrel No. 2 | | Verified | _____ |
| Safe-ready cylinder operation | 4.4.5.22.2 | | |
| Barrel No. 1 | | Verified | _____ |
| Barrel No. 2 | | Verified | _____ |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

| | Reference | Limits | Reading |
|---|---|---|---|
| Control valve operating lever | 4.4.5.22.3 | | |
| Barrel No. 1 | | No leakage, will not fire | _____ |
| Barrel No. 2 | | No leakage, will not fire | _____ |
| Ready light circuit operation | 4.4.5.22.4 | | |
| Barrel No. 1 | | Verified | _____ |
| Barrel No. 2 | | Verified | _____ |
| Breakaway plug mechanism | 4.4.5.22.5 | | |
| Barrel No. 1 | | Latched, hand unmatchable | _____ |
| Barrel No. 2 | | Latched, hand unmatchable | _____ |
| Electrical firing test | 4.4.5.22.6 | | |
| Barrel No. 1 | | One 400-milli-second voltage application per firing | _____ |
| Barrel No. 2 | | One 400-milli-second voltage application per firing | _____ |
| Observation for damage | 4.4.5.22.7 | | |
| Barrel No. 1 | | No damage | _____ |
| Barrel No. 2 | | No damage | _____ |
| * 30.2.23 Torpedo exit velocity | 4.4.5.23 | | |
| Barrels and rack properly oriented | | Verified | _____ |
| Safe-ready cylinder operation | 4.4.5.23.1 | | |
| Barrel No. 1 | | Verified | _____ |
| Barrel No. 2 | | Verified | _____ |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

| | Reference | Limits | Reading |
|---|---|---|---|
| Control valve operating lever | 4.4.5.23.2 | | |
| Barrel No. 1 | | NO leakage | _____ |
| Barrel No. 2 | | No leakage | _____ |
| Ready light circuit condition | 4.4.5.23.3 | | |
| Barrel No. 1 | | Operation per Schedule II | _____ |
| Barrel No. 2 | | Operation per Schedule II | _____ |
| Breakaway plug mechanism | 4.4.5.23.4.1 | | |
| Barrel No. 1 | | Unlatched, attached to torpedo receptacle | _____ |
| Barrel No. 2 | | unlatched, attached to torpedo receptacle | _____ |
| Breakaway plug | 4.4.5.23.4.2 | | |
| Barrel No. 1 | | Latched, hand unmatchable | _____ |
| Barrel No. 2 | | Latched, hand unmatchable | _____ |
| Breakaway plug | 4.4.5.23.4.3 | | |
| Barrel No. 1 | | NO damage after each firing | _____ |
| Barrel No. 2 | | No damage after each firing | _____ |
| Electrical firing | 4.4.5.23.5 | | |
| Barrel No. 1 | | One 400-millisecond voltage application per firing | _____ |
| Barrel No. 2 | | One 400-millisecond voltage application per firing | _____ |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

| | Reference | Limits | Reading |
|---|---|---|---|
| Manual firing | 4.4.5.23.6 | | |
| Barrel No. 1 | | 25 lbs max | _____ |
| Barrel No. 2 | | 25 lbs max | _____ |
| Observation for damage | 4.4.5.23.7 | | |
| Barrel No. 1 | | No damage | _____ |
| Barrel No. 2 | | No damage | _____ |
| Torpedoes (after launch) | | No damage | _____ |
| 30.3 Packaging, packing, and marking | | | |
| Preservation and packaging | 5.1 | Verified | _____ |
| Packing | 5.2 | Verified | _____ |
| Marking | 5.3 | Verified | _____ |



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers' vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-T-82571A(OS), Torpedo Tube, Surface Vessel, Mark 32 Mod 9

| NAME OF ORGANIZATION AND ADDRESS | CONTRACT NUMBER |
|---|---|
| | MATERIAL PROCURED UNDER A<br>☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT |

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

| SUBMITTED BY *(Printed or typed name and address - Optional)* | TELEPHONE NO. |
|---|---|
| | DATE |

**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID




OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commanding Officer
Naval Ordnance Station (502)
Standardization Division
Indian Head, Maryland 20640

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:15Z
Check the source to verify that this is the current version before use.